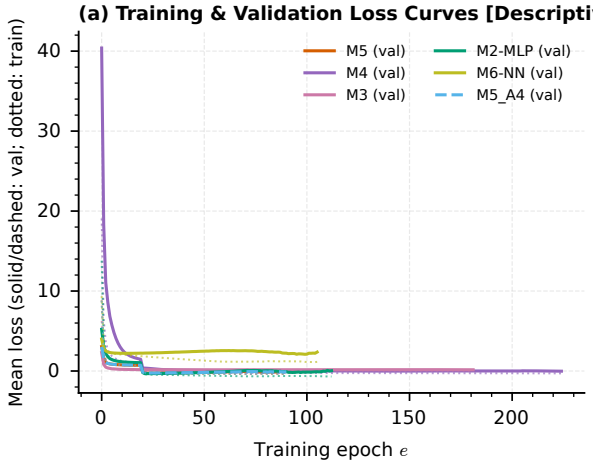
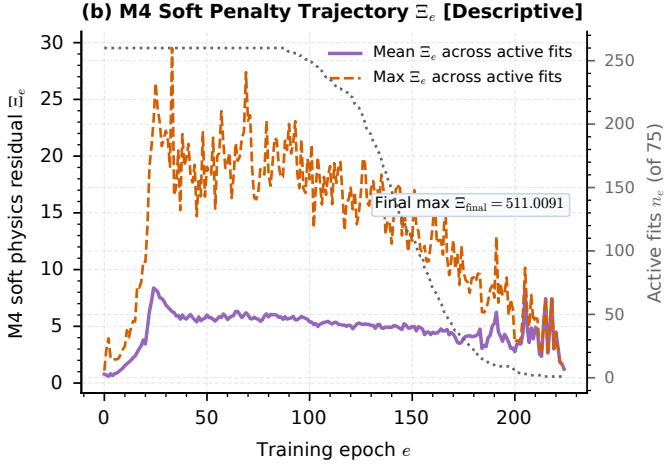
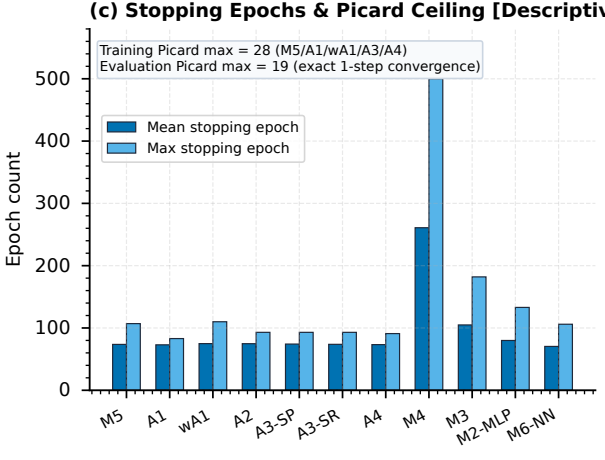
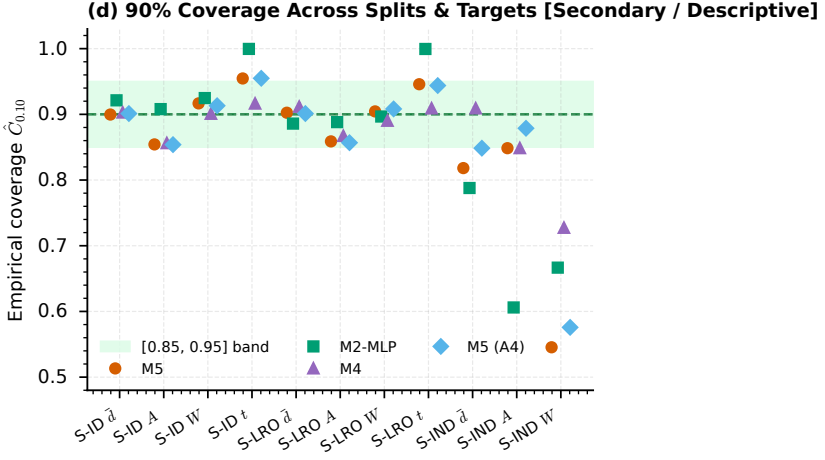

(a) Training & Validation Loss Curves [Descriptive]
M5 (val)
M4 (val)
M3 (val)
M2-MLP (val)
M6-NN (val)
M5_A4 (val)
Mean loss (solid/dashed: val; dotted: train)
Training epoch e


(b) M4 Soft Penalty Trajectory Ξ_e [Descriptive]
Mean Ξ_e across active fits
Max Ξ_e across active fits
Final max $\Xi_{\text{final}} = 511.0091$
M4 soft physics residual Ξ_e
Active fits n_e (of 75)
Training epoch e


(c) Stopping Epochs & Picard Ceiling [Descriptive]
Training Picard max = 28 (M5/A1/wA1/A3/A4)
Evaluation Picard max = 19 (exact 1-step convergence)
Mean stopping epoch
Max stopping epoch
Epoch count
M5
A1
wA1
A2
A3-SP
A3-SR
A4
M4
M3
M2-MLP
M6-NN


(d) 90% Coverage Across Splits & Targets [Secondary / Descriptive]
Empirical coverage $\hat{C}_{0.10}$
[0.85, 0.95] band
M5
M2-MLP
M4
M5 (A4)
S-ID $\bar{a}$
S-ID A
S-ID W
S-ID t
S-LRO $\bar{a}$
S-LRO A
S-LRO W
S-LRO t
S-IND $\bar{a}$
S-IND A
S-IND W